

Nadia Qazi

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EDUCATION

HARVEY MUDD COLLEGE

Bachelor of Science in Engineering with Economics Concentration

Claremont, CA

Expected May 2028

THE AMERICAN SCHOOL IN JAPAN

High School Diploma with Honors

Tokyo, JP

June 2024

RELEVANT COURSES

E80: Continuum Mechanics, E80: Experimental Engineering, E72: Applied Math for Engineers, E79: Engineering Systems, E4: Introduction to Manufacturing and Design, CS5: Introduction to Computer Science

SKILLS

Technical: Computer-Aided Design (Solidworks), Microsoft Office, Python (Intermediate), MATLAB, Arduino IDE

Languages: English, Japanese (Fluent)

PROJECTS

Autonomous Underwater Vehicle (AUV)

January 2026 - May 2026

- Built and deployed a fully autonomous underwater robot to profile light attenuation and temperature in kelp habitats at Dana Point Harbor
- Designed op-amp amplification circuits for pressure, thermistor, and RGB sensors, calibrated to Teensy microcontroller ADC ranges
- Waterproofed electronics housing using epoxy penetrators and PVC structural framing
- Validated RGB light attenuation data against Beer-Lambert law predictions across 3m depth, confirming differential wavelength decay across red, green, and blue channel

Undergraduate Researcher

January 2026 – Present

Transdermal Transport Research Lab

- Investigating the effect of mechanical skin stretching on stratum corneum permeability to improve transdermal drug delivery systems
- Conducting experiments and building test apparatus to measure transport rates across mechanically extended and hydrated skin samples

Human-Centered Adaptive Product Design Engineer

May 2025 - August 2025

Project Funded by Ben Huppee & Strauss Community Engagement Grant

- Initiated a project addressing the lack of affordable, customized assistive tools for older adults
- Designed, CAD-modeled, and 3D printed adaptive devices—including a rotating spoon, chopstick aids, and card holders—impacting 50+ seniors served by AgingNext
- Conducted user interviews and applied rapid prototyping to iterate on personalized solutions
- Managed all aspects of the project lifecycle, including research, budgeting, stakeholder coordination, and final product delivery

LEADERSHIP

TOM Fellowship

Adaptive Design at Mudd (ADAM) Co-President

August 2025 - Present

- Selected for global fellowship in inclusive design, accessibility, and community-driven innovation
- Leading 2 engineering teams weighted gardening tools and nutella seal openers
- Currently organizing fundraising events with local coffee shops to fund a switch-adaptive toy event for 30+ children with disabilities
- Trained in universal design, co-creation methods, and social impact innovation through TOM's international orientation program